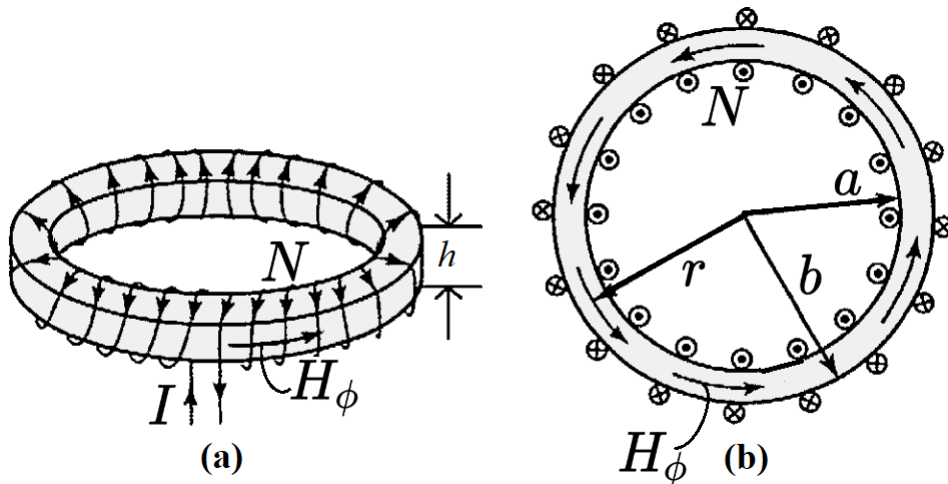


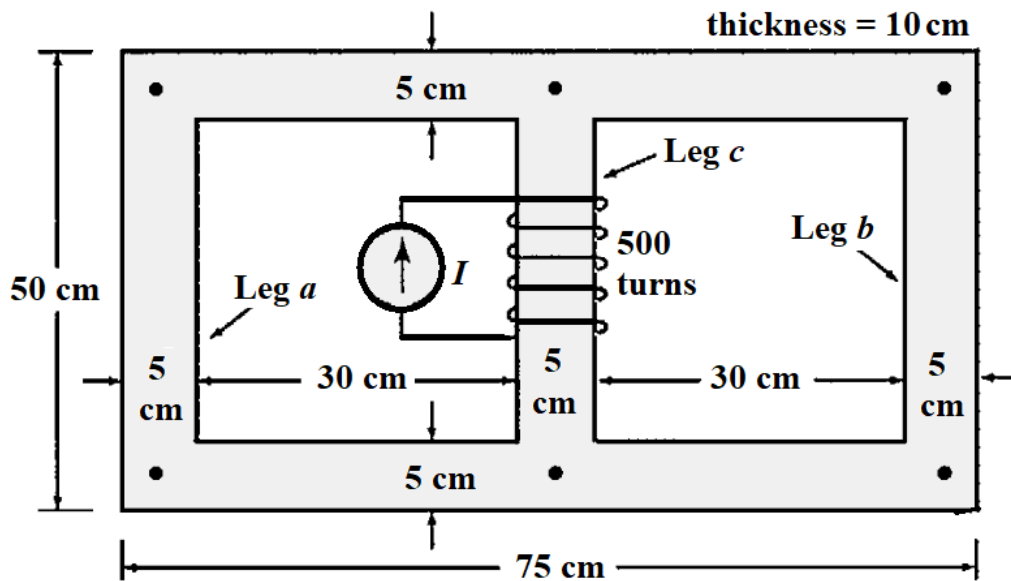


Q1: [100] A closely spaced winding with N turns is wound in the form of a ring, as shown in Figure 1. The inner and the outer radii of the ring are a and b , respectively. The height of the ring is h . If the winding carries a current of I amperes, find

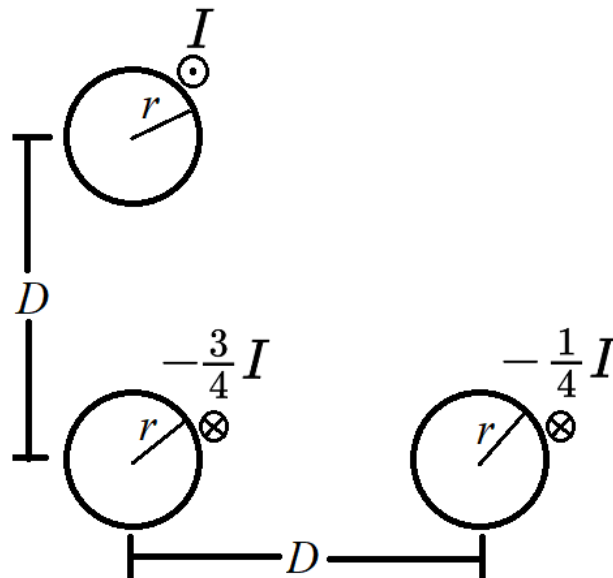
- (a) the magnetic field intensity within the ring,
- (b) the magnetic flux density,
- (c) the total flux enclosed by the ring,
- (d) the inductance, and
- (e) the total stored energy.



Q2: [50] If the 500-turn coil in Figure 2 carries a current of 2 A, determine the flux in each leg of the magnetic circuit. The relative permeability of the core is 1000.



Q3: [50] Find the per unit length inductance of the following three-wire circuit:



①

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Q1 solution.

(a) Solving $\oint \vec{H} \cdot d\vec{l} = I_{\text{enclosed}}$

Considering an Amperian path in the middle of the core at radius r , such that $a \leq r \leq b$

$$\vec{H} = H_{\phi} \hat{\phi}$$

$$d\vec{l} = r d\phi \hat{\phi}$$

$$I_{\text{enclosed}} = NI$$

$$\int_0^{2\pi} H_{\phi} r d\phi = NI$$

$$\Rightarrow H_{\phi} = \frac{NI}{2\pi r} \left[\frac{A}{m} \right] \quad a \leq r \leq b$$

(b) $\vec{B} = \mu_0 \mu_r \vec{H} \quad [Wb/m^2]$

We assume that core is a simple material. where linear relation b/w $\vec{B}$ & $\vec{H}$ exists.

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$$(c) \quad \Phi = \int \vec{B} \cdot d\vec{s}$$

$$d\vec{s} = dz dr \hat{\phi}$$

$$\vec{B} = \mu_0 \mu_r \frac{NI}{2\pi r} \hat{\phi}$$

$$\Phi = \frac{\mu_0 \mu_r}{2\pi} NI \int_{z=0}^h \int_{r=a}^b \frac{1}{r} dz dr$$

$$\Phi = \frac{\mu_0 \mu_r NI h}{2\pi} \ln(b/a) \quad [\text{wb}]$$

$$(d) \quad L = \frac{\lambda}{I} = \frac{\text{Flux linkage}}{\text{Source Current.}}$$

$$\lambda = N \Phi$$

Because the enclosed current is N times the source current.

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$$L = \frac{N\Phi}{I} = \frac{\mu_0 \mu_r N^2 h \ln(b/a)}{2\pi} [H]$$

(e)

$$W_m = \frac{1}{2} \int_V \vec{B} \cdot \vec{H} \, dv$$

$$= \frac{1}{2} \mu_0 \mu_r \frac{N^2 I^2}{4\pi^2} \iiint \frac{1}{r^2} r \, dr \, d\phi \, dz$$

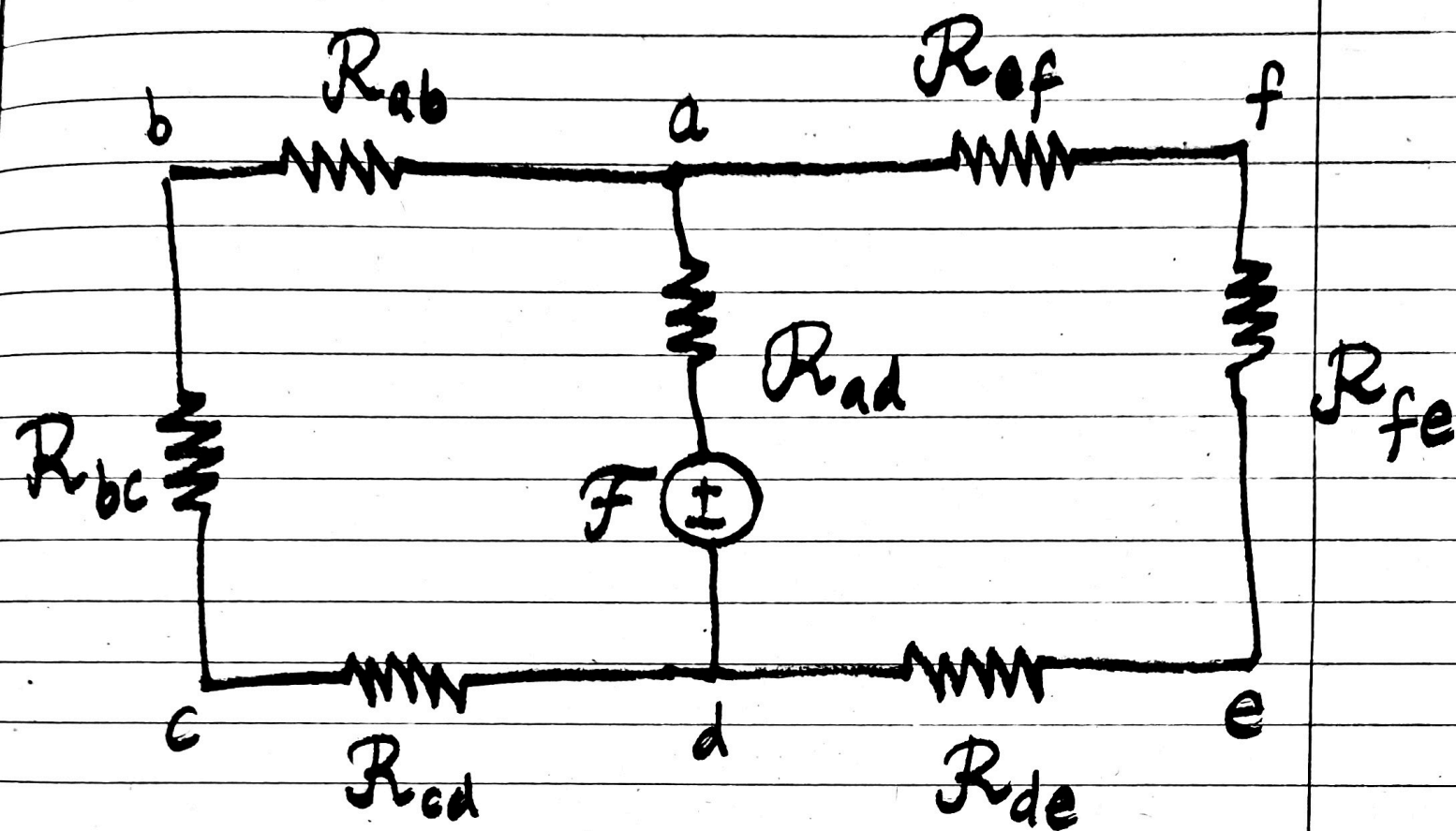
$$= \frac{\mu_0 \mu_r N^2 I^2 h \ln(b/a)}{4\pi} [\text{Joules}]$$

④

Q2.

$$\mu_r = 1000$$

$$\mu_0 = 4\pi \times 10^{-7}$$



Note that

$$R_{ab} = R_{cd} = R_{de} = R_{ef} =: R_1$$

$$l = \text{Length} = \frac{75 - 2.5 - 2.5}{2} = 35 \text{ cm}$$

$$S = \text{Area} = 10 \times 10^{-2} \times 5 \times 10^{-2} = 5 \times 10^{-3} \text{ m}^2$$

$$R_1 = \frac{l}{\mu_0 \mu_r S} = 5.5704 \times 10^4$$

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$$R_2 := R_{bc} = R_{ad} = R_{fe}$$

$$l = 50 - \frac{5}{2} - \frac{5}{2} = 45 \times 10^{-2} \text{ m}$$

$$S = 50 \times 10^{-4} = 5 \times 10^{-3} \text{ m}^2$$

$$R_2 = \frac{l}{\mu_0 \mu_r S} = 7.162 \times 10^4$$

$$\text{Thevenin reluctance} = (2R_1 + R_2) \parallel (2R_1 + R_2) + R_2$$

$$R_{th} = (R_1 + \frac{1}{2} R_2) + R_2$$

$$= 1.6313 \times 10^5$$

$$\psi_{total} = \psi_{legc} = \frac{NI}{R_{th}} = 0.0061 \text{ Wb}$$

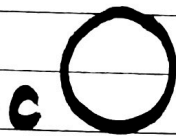
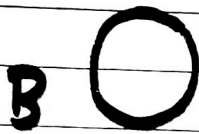
$$\psi_{lega} = \psi_{legb} = \frac{0.0061}{2} = 0.0031 \text{ Wb.}$$

⑥

$$r' = 0.7788r$$

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Q3.



$$\lambda_A = 2 \times 10^{-7} \left(I \ln\left(\frac{1}{r'}\right) - \frac{3I}{4} \ln\left(\frac{1}{D}\right) - \frac{I}{4} \ln\left(\frac{1}{\sqrt{2}D}\right) \right)$$

$$\lambda_A = 2 \times 10^{-7} I \left(\ln\left(\frac{1}{r'}\right) + \ln\left(D^{3/4}\right) + \ln\left(D^{1/4} 2^{1/8}\right) \right)$$

$$= 2 \times 10^{-7} I \left(\ln\left(\frac{D}{r'} 2^{1/8}\right) \right)$$

$$L_A = \frac{\lambda_A}{I_A} = \frac{\lambda_A}{I} = 2 \times 10^{-7} \ln\left(\frac{2^{1/8} D}{r'}\right) \quad [\text{H/m}]$$

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$$\begin{aligned}\lambda_B &= 2 \times 10^{-7} \left[I \ln\left(\frac{1}{D}\right) - \frac{3I}{4} \ln\left(\frac{1}{r'}\right) \right. \\ &\quad \left. - \frac{I}{4} \ln\left(\frac{1}{D}\right) \right] \\ &= 2 \times 10^{-7} I \left(\ln\left(\frac{1}{D}\right) + \ln\left(r'^{3/4}\right) \right. \\ &\quad \left. + \ln\left(D^{1/4}\right) \right)\end{aligned}$$

$$= 2 \times 10^{-7} I \ln\left(\frac{r'^{3/4}}{D^{3/4}}\right)$$

$$= -\frac{6}{4} \times 10^{-7} I \ln\left(\frac{D}{r'}\right)$$

$$L_B = \frac{\lambda_B}{\left(-\frac{3I}{4}\right)} = 2 \times 10^{-7} \ln\left(\frac{D}{r'}\right)$$

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$$\lambda_c = 2 \times 10^{-7} \left[-\frac{I}{4} \ln\left(\frac{1}{r'}\right) - \frac{3I}{4} \ln\left(\frac{1}{D}\right) + I \ln\left(\frac{1}{\sqrt{2}D}\right) \right]$$

$$= 2 \times 10^{-7} I \left[\ln\left(r'^{\frac{1}{4}}\right) + \ln\left(D^{\frac{3}{4}}\right) + \ln\left(\frac{1}{\sqrt{2}D}\right) \right]$$

$$= 2 \times 10^{-7} I \ln\left(\frac{r'^{\frac{1}{4}}}{\sqrt{2} D^{\frac{1}{4}}}\right)$$

$$= \frac{1 \times 10^{-7}}{2} I \ln\left(\frac{r'}{D}\right) = -\frac{1 \times 10^{-7}}{2} I \ln\left(\frac{4D}{r'}\right)$$

$$L_c = \frac{\lambda_c}{\left(-\frac{I}{4}\right)} = 2 \times 10^{-7} \ln\left(\frac{4D}{r'}\right) \quad [H/m]$$